2016.0 RANGE ROVER (LG), 501-10

SEATING – STANDARD WHEEL BASE

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Seating system, refer to the relevant Description and Operation section in the workshop manual. REFER to: Seats (501-10 Seating - Short Wheelbase, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Seat runners ▪ Seat frames	▪ Fuses ▪ Wiring harnesses and connectors ▪ Driver seat module ▪ Passenger seat module ▪ Body control module ▪ Touch screen ▪ Seat movement switch(es) ▪ Seat heater switch(es) ▪ Seat motor(s)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SEAT MOVEMENT INHIBITION CHECKS

If the customer is reporting issues with seat movements, first consult the table below to check if the seat movement functions may have been inhibited. If the issue cannot be rectified, proceed to the pinpoint tests below:

SYMPTOM	REASON FOR INHIBIT	HOW TO IDENTIFY FAULT SPECIFICS	RECTIFICATION	AFFECTED SEAT ROW	FURTHER DESCRIPTION
Seat movement not operating from switch press	▪ Seat movement motor overheat	▪ Only the affected motor will be disabled ▪ Under current DTC may be logged	▪ Wait for at least 5 minutes to allow motor to cool ▪ Perform ignition cycle	▪ 1st Row - L462, L405 & L494 ▪ 2nd Row - L405 Business Class Seating only	▪ n/a
Rear seat movement not operating from switch press	▪ Seat module software has activated seat movement motor thermal protection	▪ Only the affected motor will be disabled	▪ Wait for at least 2 minutes to allow motor to cool	▪ 2nd Row - L462, L405 60/40 Seating only ▪ 3rd Row - L462, L494 only	▪ n/a

Rear seats not folding when using load space switches	▪ Tailgate is not open	▪ Check if the tailgate is closed	▪ Open tailgate	▪ 2nd Row - L462, L405 60/40 Seating only ▪ 3rd Row - L462, L494 only	▪ n/a
	▪ Power management time out	▪ Load space lighting is not illuminated	▪ Close and open tailgate ▪ Press unlock button ▪ Turn on ignition	▪ 2nd Row - L462, L405 60/40 Seating only ▪ 3rd Row - L462, L494 only	▪ n/a
Seat fold not operating from C pillar switch	▪ Adjacent door closed	▪ Check if adjacent door is open	▪ Open adjacent door	▪ 3rd Row - L462, L494 only	▪ n/a
	▪ Power management time out	▪ The vehicle has been left without any interaction	▪ Press unlock button ▪ Turn on ignition	▪ 3rd Row - L462, L494 only	▪ n/a
Rear seat movement not operating from rear seat switch pack	▪ Rear window isolate (child lock) switch on	▪ Check if the rear window isolate (child lock) button telltale LED is illuminated	▪ Turn off the rear window isolate (child lock) switch	▪ 2nd Row - L462, L405 only	▪ n/a
Seat movement	▪ Seat controls not working	▪ Only the affected	▪ Turn ignition OFF, then ON	▪ 1st Row - L462,	▪ Fault codes (logged in

	not operating from switch press	(seat module software has activated circuit protection)	motor will be disabled	■ Check for related DTCs and repair any circuit faults	L405 & L494 ■ 2nd Row - L462, L405 only ■ 3rd Row - L462, L494 only	seat module) preventing seat movement. Check for 'Circuit Current Above/Below Threshold' DTCs and refer to the relevant DTC index
	Seat movement 'inching', not smooth operation (starts & then stops)	■ Seat movement issue – "inching" movement	■ Only the affected motor will be inching	■ Check for related DTCs and refer to the relevant DTC index	■ 1st Row - L462, L405 & L494 ■ 2nd Row - L462, L405 only	■ Fault codes (logged in seat module) preventing smooth seat movement. Check for 'Speed Position Sensor - No Signal' DTCs and refer to the relevant DTC index
	Driver seat not returning to memory position (when driving)	■ -	■ -	■ -	■ 1st Row driver seat only - L462, L405 & L494	■ Once a vehicle threshold speed (5 kph) is detected, automatic seat movement to memory position is disabled and on-going return to memory position is stopped
	Seat controls not working	■ Seat controls not working (seat module	■ -	■ Check for battery and/or charging faults and	■ 1st Row - L462, L405 & L494	■ If battery's voltage output decreases

(minimum voltage protection)	software has disabled operation)		rectify as required	▪ 2nd Row - L462, L405 only ▪ 3rd Row - L462, L494 only	below minimum allowable value of 9 volts, the seat control module stops operating
Easy Entry/Exit - Drivers seat not returning to original driving position	▪ No automated seat movement is permitted with vehicle moving	▪ -	▪ Do not drive the vehicle whilst the seat/steering column is moving to required driving position	▪ 1st Row driver seat only - L462, L405 & L494	▪ The function of easy entry will only complete and return the seat and column to the required position if the vehicle speed remains lower than a permitted vehicle threshold speed (5 kph)

SYMPTOM CHARTS

Electrical Adjustment - Front Seats

SYMPTOM	POSSIBLE CAUSES	ACTION
Front seat fore/aft movement not functioning	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **A.**
Front seat excessive fore/aft free play	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **B.**
Front seat fore/aft movement noisy	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **C.**
Front seat height, tilt and/or seat	▪ Carry out the pinpoint test	GO to

extension motor movement not functioning	associated to this symptom	Pinpoint Test **D.**
Front seat height, tilt and/or extension movement noisy	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **E.**

Heading and Cooling - Front Seats

SYMPTOM	POSSIBLE CAUSES	ACTION
Heating And Cooling - Inoperative	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **F.**
Heating And Cooling - Noisy operation	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **G.**
Heating And Cooling - Poor heat or cool efficiency	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **H.**
Heating And Cooling - Heat or cool operation slow	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **I.**
Heating And Cooling - Intermittent operation	▪ Carry out the pinpoint test associated to this symptom	GO to Pinpoint Test **J.**

PINPOINT TESTS

WARNING:

When carrying out the following steps, stand clear of all moving parts and ensure link harness is routed accordingly

PINPOINT TEST A : FRONT SEAT FORE/AFT MOVEMENT NOT FUNCTIONING	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS

A1: CHECK FOR FRONT SEAT FORWARD-REARWARD SEAT MOTOR OPERATION

	WARNING: Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual **1** Set ignition status to ' ON'
	2 From the switch pack, operate the front seat forward-rearward seat motor switch and listen for evidence of the motor operating
	Does the motor operate? **Yes** GO to A2. **No** GO to A3.

A2: CHECK FRONT SEAT FORWARD-REARWARD SEAT MOTOR DRIVE BAR

	1 Check front seat drive bar for correct installation and condition
	Is the front seat drive bar correctly installed and in a serviceable condition? **Yes** Re-check for correct front seat forward-rearward movement. Remove seat to allow for further investigation if required **No** Correctly install front seat forward-rearward seat motor drive bar, or replace if required

A3: CHECK FRONT SEAT FORWARD-REARWARD SEAT MOTOR

	1 Set ignition status to 'OFF'
	2 Disconnect front seat forward-rearward seat motor connector
	NOTE: It may be that the seat has been driven to the limit of travel along the relevant axis, and when the link harness is connected, the seat will remain in the same position. If this is the case, a jolt may be felt from the motor. To confirm the motor operation, swap the link harness to alternate motor pin connections and the seat should travel in the opposite direction **3** Using a locally made fused link harness and power supply, connect power and ground to forward-rearward seat motor

BATTERY POSITIVE TERMINAL	BATTERY NEGATIVE TERMINAL
forward-rearward seat motor pin 1	forward-rearward seat motor pin 2

	Does the motor operate? **Yes** Using manufacturer approved diagnostic system, check for related (DTCs and carry out the repair operations specified. Alternatively, refer to the electrical circuit diagrams and check front seat forward-rearward seat motor circuits **No** Replace front seat forward-rearward seat motor. Refer to relevant section of workshop manual

PINPOINT TEST B : FRONT SEAT EXCESSIVE FORWARD-REARWARD FREE PLAY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: CHECK FRONT SEAT FOR EXCESSIVE FORWARD-REARWARD FREE PLAY	
	WARNING: Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual 1 Check all accessible front seat frame fixings are installed and to the correct torque
	Are all accessible front seat frame fixings installed and to the correct torque? **Yes** GO to B2. **No** Install and tighten all accessible front seat frame fixings to correct torque and re-check for excessive free play
B2: COMPARE THE FRONT SEAT FORWARD-REARWARD FREE PLAY AGAINST A SIMILAR SEAT	
	1 Compare the front seat forward-rearward free play against a similar seat
	Is the front seat forward-rearward free play excessive when compared to a similar seat? **Yes**

	GO to B3. **No** The front seat frame is operating correctly. Submit Electronic Product Quality Report (EPQR) with any further query
B3: CHECK REMAINING FRONT SEAT FRAME FIXINGS	
	1 Remove front seat and/or any seat covers/trim to allow access to check remaining front seat frame fixings are all installed and to the correct torque
	Are all remaining front seat frame fixings installed and to the correct torque? **Yes** Replace front seat frame. Refer to the relevant section of the workshop manual **No** Install and tighten all remaining front seat frame fixings to correct torque and re-check for excessive free play

PINPOINT TEST C : FRONT SEAT FORWARD-REARWARD MOVEMENT NOISY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: COMPARE FRONT SEAT FORWARD-REARWARD MOVEMENT NOISE TO OTHER FRONT SEAT	
	WARNING: Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual 1 Compare the front seat forward-rearward movement noise to other front seat
	Is the front seat forward-rearward movement noise excessive when compared to other front seat? **Yes** GO to C2. **No** GO to C3.
C2: COMPARE FRONT SEAT FORWARD-REARWARD MOVEMENT NOISE TO FRONT SEAT IN OTHER VEHICLE	
	1 Compare the front seat forward-rearward movement noise to front seat in other vehicle
	Is the front seat forward-rearward movement noise excessive when

	compared to front seat in other vehicle? **Yes** GO to C3. **No** The front seat frame is operating correctly. Submit Electronic Product Quality Report (EPQR) with any further query

C3: CHECK FOR DEBRIS OBSTRUCTING SEAT MOVEMENT

	1 Check for debris obstructing seat movement
	Is the front seat forward-rearward movement obstructed by debris? **Yes** Remove obstruction and re-check for noisy forward-rearward seat movement **No** GO to C4.

C4: RE-ALIGN FRONT SEAT FRAME

	1 Loosen front seat frame fixings
	2 Set ignition status to 'ON'
	3 Using the front seat switch pack drive the front seat fully forward then fully rearward
	4 Tighten front seat frame fixings to the correct torque
	5 Re-check for noisy seat movement
	Is the front seat forward-rearward movement still noisy? **Yes** GO to C5. **No** The front seat frame is now operating correctly

C5: CHECK FRONT SEAT FORWARD-REARWARD SEAT MOTOR DRIVE BAR

	1 Check front seat drive bar for correct installation and condition
	Is the front seat drive bar correctly installed and in a serviceable condition? **Yes** Replace front seat forward-rearward seat motor. Refer to relevant section of workshop manual **No** Correctly install front seat forward-rearward seat motor drive bar, or replace if required

WARNINGS:

- Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual
- When carrying out the following steps, stand clear of all moving parts and ensure link harness is routed accordingly

PINPOINT TEST D : FRONT SEAT HEIGHT, TILT AND/OR SEAT EXTENSION MOTOR MOVEMENT NOT FUNCTIONING	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: CHECK FRONT SEAT HEIGHT, TILT OR EXTENSION MOTOR	
	1 Set ignition status to ' OFF'
	2 Disconnect front seat height, tilt or extension motor connector
	NOTE: It may be that the seat has been driven to the limit of travel along the relevant axis, and when the link harness is connected, the seat will remain in the same position. If this is the case, a jolt may be felt from the motor. To confirm the motor operation, swap the link harness to alternate motor pin connections and the seat should travel in the opposite direction **3** Using a locally made fused link harness and power supply, connect power and ground to relevant motor
	Does the motor operate? **Yes** Using manufacturer approved diagnostic system, check for related DTCs and carry out the repair operations specified. Alternatively, refer to the electrical circuit diagrams and check relevant motor circuits **No**

BATTERY POSITIVE TERMINAL	BATTERY NEGATIVE TERMINAL
motor pin 1	motor pin 2

	Replace the relevant motor. Refer to relevant section of workshop manual

PINPOINT TEST E : FRONT SEAT HEIGHT, TILT AND/OR EXTENSION MOVEMENT NOISY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: COMPARE THE HEIGHT, TILT OR EXTENSION MOVEMENT NOISE WITH THE OTHER FRONT SEAT	
	WARNING: Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual **1** Compare the front seat movement noise to other front seat
	Is the front seat height, tilt or extension movement noise excessive when compared to other front seat? **Yes** **GO to E2**. **No** **GO to E3**.
E2: COMPARE FRONT SEAT HEIGHT, TILT OR EXTENSION MOVEMENT NOISE TO FRONT SEAT IN OTHER VEHICLE	
	1 Compare the front seat height, tilt or extension movement noise to front seat in other vehicle
	Is the front seat height, tilt or extension movement noise excessive when compared to front seat in other vehicle? **Yes** **GO to E3**. **No** The front seat frame is operating correctly. Submit Electronic Product Quality Report (EPQR) with any further query
E3: CHECK FOR DEBRIS OBSTRUCTING SEAT MOVEMENT	
	1 Check for debris obstructing seat movement
	Is the front seat height, tilt or extension movement obstructed by debris? **Yes** Remove obstruction and re-check for noisy height, tilt or extension seat movement. If still noisy GO to Pinpoint Test **G.** **No** **GO to E4**.

E4: CHECK FOR HEIGHT, TILT OR EXTENSION MOVEMENT MECHANISM LUBRICATION	
	1 Check and apply manufacturer approved lubrication to seat height, tilt or extension movement mechanism and retest for noise
	Is the front seat height, tilt or extension noise still apparent? **Yes** Replace the relevant motor. Refer to relevant section of workshop manual **No** The front seat height, tilt or extension motor is operating correctly

WARNING:

Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual

NOTES:

- The climate controlled seat's (heat and cool) functions will only operate when the vehicle's engine is running
- If a fault is identified and repaired, check for correct operation of the driver and front-passenger climate controlled seat (heat and cool) functions

PINPOINT TEST F : HEATER AND COOLING - INOPERATIVE	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: CHECK FOR STORED DTC	
	1 Check the seat climate control module for stored DTCs
	Are there any stored DTCs? **Yes** For information on stored DTCs refer to the information in the seat climate control module DTC index. **No** GO to F2.

F2: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function of the backrest
	Does the backrest heat and cool function operate correctly? **Yes** GO to F3. **No** GO to F5.

F3: DRIVER OR FRONT PASSENGER SEAT CUSHION CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function at the cushion
	Does the cushion heat and cool function operate correctly? **Yes** If there are no faults evident, verify the customer concern **No** GO to F4.

F4: CUSHION BELLOWS	
	1 Check the condition of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required

F5: BACKREST BELLOWS DUCT	
	1 Check the security of the bellows duct
	Is the bellows duct correctly installed? **Yes** GO to F6. **No** Securely reconnect the bellows duct

F6: BACKREST BELLOWS	
	1 Check the security of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** Suspect an internal fault with the climate controlled seat. Replace as required. Follow the link below to check operation of the driver or front passenger seat cushion climate assemblyGO to F2.

WARNING:

Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual

NOTES:

- The climate controlled seat's (heat and cool) functions will only operate when the vehicle's engine is running
- If a fault is identified and repaired, check for correct operation of the driver and front-passenger climate controlled seat (heat and cool) functions

PINPOINT TEST G : HEATING AND COOLING - NOISY OPERATION	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: CHECK FOR STORED DTC	
	1 Check the seat climate control module for stored DTCs
	Are there any stored DTCs? **Yes** For information on stored DTCs refer to the information in the seat climate control module DTC index **No** GO to G2.
G2: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - NOISE	
	1 There is a known issue which affects a limited number of vehicles where under acceleration of the vehicle the occupant of seat can deform the suspension mat (snake wire) which then presses on the casing of the seat backrest climate assembly, this causes the casing to distort and the noise issue to occur. If contact has occurred there will be a witness mark on the casing. Carry out visual inspection of the seat backrest climate assembly that has been identified as noisy by the customer
	Is there a witness mark on the casing?

	Yes Contact the local in market support for further information **No** **GO to G3**.

G3: DRIVER OR FRONT PASSENGER SEAT CLIMATE ASSEMBLY - NOISE 2	
	1 Operate the heat and cool function of the seat. Listen for a 'WAH WAH' noise (the fan speeds up and slows down repeatedly)
	Does the fan speeds up and slow down repeatedly? **Yes** Confirm the latest Strategy and Calibration software is installed, using the manufacturer approved diagnostic system carry out the new seat climate control module application and update the seat climate control module software if requiredIf the noise is still evident after software update follow link below**GO to G4**. **No** **GO to G4**.

G4: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - NOISE LEVEL COMPARISON	
	1 Compare the noise level of the suspect climate seat assembly to another climate seat assembly
	Is the noise level equal between the two seats? **Yes** The noise level is standard and comparable to the design intent **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required

WARNING:

Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual

NOTES:

- The climate controlled seat's (heat and cool) functions will only operate when the vehicle's engine is running
- If a fault is identified and repaired, check for correct operation of the driver and front-passenger climate controlled seat (heat and cool) functions

PINPOINT TEST H : HEATING AND COOLING - POOR HEAT OR COOL EFFICIENCY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: CHECK FOR STORED DTC	
	1 Check the seat climate control module for stored DTCs
	Are there any stored DTCs? **Yes** For information on stored DTCs refer to the information in the seat climate control module DTC index **No** **GO to H2**.
H2: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function of the backrest
	Does the backrest heat and cool function operate correctly? **Yes** **GO to H3**. **No** **GO to H5**.
H3: DRIVER OR FRONT PASSENGER SEAT CUSHION CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function at the cushion
	Does the cushion heat and cool function operate correctly? **Yes** If there are no faults evident, verify the customer concern **No** **GO to H4**.
H4: CUSHION BELLOWS	

	1 Check the condition of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** **GO to H7**.

H5: BACKREST BELLOWS DUCT

	1 Check the security of the bellows duct
	Is the bellows duct correctly installed? **Yes** **GO to H6**. **No** Securely reconnect the bellows duct

H6: BACKREST BELLOWS

	1 Check the security of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required. Follow the link below to check operation of the driver or front passenger seat cushion climate assembly.**GO to H2**.

H7: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - EFFICIENCY COMPARISON

	1 Compare the efficiency of the suspect climate seat assembly to another climate seat assembly
	Is the efficiency equal between the two seats? **Yes** The efficiency is standard and comparable to the design intent **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required

WARNING:

Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual

NOTES:

- The climate controlled seat's (heat and cool) functions will only operate when the vehicle's engine is running
- If a fault is identified and repaired, check for correct operation of the driver and front-passenger climate controlled seat (heat and cool) functions

PINPOINT TEST I : HEATING AND COOLING - HEAT OR COOL OPERATION SLOW	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: CHECK FOR STORED DTC	
	1 Check the seat climate control module for stored DTCs
	Are there any stored DTCs? **Yes** For information on stored DTCs refer to the information in the seat climate control module DTC index **No** GO to I2.
I2: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function of the backrest?
	Does the backrest heat and cool function operate correctly? **Yes** GO to I3. **No** GO to I5.
I3: DRIVER OR FRONT PASSENGER SEAT CUSHION CLIMATE ASSEMBLY - FUNCTIONALITY CHECK	
	1 Check the heat and cool function at the cushion
	Does the cushion heat and cool function operate correctly? **Yes** If there are no faults evident, verify the customer concern **No** GO to I4.
I4: CUSHION BELLOWS	

	1 Check the condition of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** GO to I7.

I5: BACKREST BELLOWS DUCT

	1 Check the security of the bellows duct
	Is the bellows duct correctly installed? **Yes** GO to I6. **No** Securely reconnect the bellows duct

I6: BACKREST BELLOWS

	1 Check the security of the bellows
	Are the bellows obstructed, have they collapsed? **Yes** Remove the obstruction or replace collapsed bellows **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required. Follow the link below to check operation of the driver or front passenger seat cushion climate assemblyGO to I2.

I7: DRIVER OR FRONT PASSENGER SEAT BACKREST CLIMATE ASSEMBLY - OPERATION COMPARISON

	1 Compare the operation of the suspect climate seat assembly to another climate seat assembly
	Is the operation equal between the two seats? **Yes** The operation is standard and comparable to the design intent **No** Suspect an internal fault with the climate controlled seat assembly. Replace as required

WARNING:

Before work is carried out, make the air bag supplemental restraint system safe. For additional information, refer to Standard Workshop Practices section of workshop manual

NOTES:

- The climate controlled seat's (heat and cool) functions will only operate when the vehicle's engine is running
- If a fault is identified and repaired, check for correct operation of the driver and front-passenger climate controlled seat (heat and cool) functions

PINPOINT TEST J : HEATING AND COOLING - INTERMITTENT OPERATION	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
J1: CHECK FOR STORED DTC	
	1 Check the seat climate control module for stored DTCs
	Are there any stored DTCs? **Yes** For information on stored DTCs refer to the information in the seat climate control module DTC index **No** GO to J2.
J2: CABLE CONNECTION CHECK	
	1 Identify from customer concern or stored DTCs which Seat Climate Assembly is operating intermittently and confirm the harness connector is fully connected
	Is the harness connector is fully connected? **Yes** Suspect an internal fault with the climate controlled seat assembly. Replace as required **No** Disconnect the harness connector, inspect connector and terminals for damage. Repair or replace if required. Reconnect connector and check operation

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

Diagnostic Trouble Code Index - DTC: Seat Module - Driver / Passenger / Rear - Left / Right (Description and Operation),
Diagnostic Trouble Code Index - DTC: Automatic Temperature Control Module (ATCM) (Description and Operation).